Origins and Evolution of Language
Week 7 tutorial
Tutor notes

Comments for tutors are in italics.

As usual week, two aims for today:

1. *Read and talk about an interesting paper from one of our own.*
2. *More generally, practice reading, summarising, and evaluating research papers, with a bit of encouragement from questions are intended to help think critically about the paper's methods and conclusions. This paper is particularly interesting because it models really careful evaluation and critique of other papers in the literature.*

One additional note: you could easily do 2 tutorials on this paper, it's very rich. I decided to skip the Fitch & Hauser stuff entirely and focus on Kanzi, which comes with a fun video and is less dry than the AnBn stuff.

This week's workshop picks up on a topics we discussed last week, on the grammar learning capacities of non-humans. In preparation for this workshop you will read and prepare to discuss parts of Truswell (2017), which is a short critique of a well-known monkey artificial grammar learning experiment (section 2, which we won't focus on and which you can feel free to skip), plus an interesting analysis of the comprehension abilities of a language-trained ape (section 3, and section 4 for a comparison to human children). The author, Prof. Rob Truswell, is based here in Edinburgh, you might know him.

You should read sections 1, 3, 4 and 5 of the Truswell paper. If you choose to read section 2, there is some additional context below, but section 2 is not required. Sections 3 onwards provide an evaluation of Savage-Rumbaugh et al.'s (1993) data on Kanzi's interpretation of spoken English commands, with a focus on what they tell us about Kanzi's knowledge of English grammar, specifically his sensitivity to linear order and hierarchical structure. Kanzi is a male bonobo (a species closely related to common chimpanzees) who was raised by humans in a research lab in quite unusual circumstances. You can see a video of Kanzi performing a similar task (or maybe this is some of the actual data?) on youtube, worth watching all the way through once you have read the paper, applying Truswell's eye to Kanzi's actions!

The second part of the paper is an evaluation of Savage-Rumbaugh et al.'s (1993) data on Kanzi's interpretation of spoken English commands, with a focus on what they tell us about Kanzi's knowledge of English grammar, specifically his sensitivity to linear order and hierarchical structure. Kanzi is a male bonobo (a species closely related to common chimpanzees) who was raised by humans in a research lab in quite unusual circumstances. You can see a video of Kanzi performing a similar task (or maybe this is some of the actual data?) on youtube, worth watching all the way through once you have read the paper, applying Truswell's eye to Kanzi's actions!

There's some stuff in there you could solve with a semantic soup strategy – "cut onions knife", "put pine-needles refrigerator", "carry TV outdoors" can only be interpreted one way –

but some stuff that shows sensitivity to linear order ("put soap water", "pour coke water") – might be worth watching and discussing these when you get to that section of the tutorial.

Questions for discussion:

- What is dendrophilia? What is dendrophobia?

The idea is that dendrophiles will be predisposed to infer hierarchical structure over stringsets (i.e. they might run ahead of the evidence and hallucinate hierarchical structure from inconclusive evidence) whereas dendrophobes will be reluctant to do so (and might not do so even given very convincing evidence that they should). You could relate in a general way this to the learning bias we saw in last week's tutorials on zebra finch song – the learning bias we are talking about here is different (what kind of abstract grammar you infer that is consistent with your data, rather than preference for e.g. shorter notes in the middle and better rhythm where you depart from your training data), but the general idea that you can have expectations about how your communication system will work is the same.

- What is "semantic soup" as an interpretation strategy, and why does Truswell consider it? What evidence is there against Kanzi applying a semantic soup strategy?

In a semantic soup strategy "the meanings of the individual content words are formed into a coherent action description in whatever way they fit best, without attention to any syntactic information in the signal. " Truswell considers it because he is essentially applying Morgan's Cannon – look for the simplest possible explanation to account for the observed behaviour, and here we assume that semantic soup is a simpler interpretative strategy because it involves less grammar (but on the other hand, it does involve more interpretation!). The main evidence against this in the Kanzi data is his sensitivity to linear order – e.g. sentences like "pour coke water" in the video above seem to be interpreted fairly reliably.

- What is structure sensitivity, and what evidence is there that Kanzi does not exhibit structure sensitivity? What evidence is there that human infants exhibit structure sensitivity? How does this relate to dendrophilia and dendrophobia?

Structure sensitivity involves being sensitive to factors beyond linear order, understanding that the word string is composed of constituents: in the case of this paper the critical example are conjoined NPs, where you have to realise that the action given in the verb should be applied to both nouns in the conjoined NP, i.e. they form a complex argument to the verb. Inferring these abstract hierarchical groupings indicates dendrophilia, or at least the capacity to overcome dendrophobia, because they involve inferring hierarchical structure over strings.

- More generally, what do you think about the prospects of this kind of experimental data (like that from Savage-Rumbaugh et al., or the animal artificial grammar learning experiments we discussed in the lectures) to tell us something about the evolution of the human capacity for language? Are there particular studies you'd like to see run? If so, what studies? If not, why not?

This is one for general discussion. Personally I think this stuff is very promising, but challenging, and to be done well it really needs a keen eye on the design of the target languages (basically trying to minimise alternative cues that could be used), the test sets (including demonstrating generalisation beyond the training examples), and the interpretation on the results. For me the best prospects here are well-controlled lab studies, rather than the relatively free-and-easy procedure often adopted in old-school animal language learning experiments, like the one involving Kanzi. I think its most useful possible finding is helping us to narrow down exactly what the difference is between humans and other animals in ability/propensity to infer particular grammar types. Are we completely off the charts relative to our closest relatives, or can we see some continuity in these abilities? Have other species (e.g. birds) independently evolved grammar-learning abilities that are closer to humans, and if so what shared selection pressures might have lead to this convergent evolution?